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Oefeningenreeks 4A nr. 43.2

$$I = \int \frac{x^2 + 7x + 1}{(x + 2)^2(2x + 1)} dx$$

$$\frac{x^2 + 7x + 1}{(x + 2)^2(2x + 1)} = \frac{A}{2x + 1} + \frac{B}{(x + 2)^2} + \frac{C}{x + 2}$$

$$A = \left. \frac{x^2 + 7x + 1}{(x + 2)^2} \right|_{x=-\frac{1}{2}} = -1$$

$$B = \left. \frac{x^2 + 7x + 1}{2x + 1} \right|_{x=-2} = 3$$

$$\begin{aligned} & \left(\frac{x^2 + 7x + 1}{(x + 2)^2(2x + 1)} - \frac{3}{(x + 2)^2} \right) (x + 2) \\ &= \left(\frac{(x^2 + 7x + 1) - 3(2x + 1)}{(x + 2)^2(2x + 1)} \right) (x + 2) = \frac{x^2 + x - 2}{(x + 2)(2x + 1)} \\ &= \frac{(x + 2)(x - 1)}{(x + 2)(2x + 1)} = \frac{x - 1}{2x + 1} \end{aligned}$$

$$\Rightarrow C = \left. \frac{x - 1}{2x + 1} \right|_{x=-2} = 1$$

$$\Rightarrow I = \int \frac{-dx}{2x + 1} + \int \frac{3dx}{(x + 2)^2} + \int \frac{dx}{x + 2}$$

$$= - \int \frac{dx}{2x + 1} + 3 \int \frac{dx}{(x + 2)^2} + \int \frac{dx}{x + 2}$$

$$= -\frac{\ln|2x + 1|}{2} - \frac{3}{x + 2} + \ln|x + 2| + C$$

References